



- [Home](#)
- [Wooden Boat Plans](#)
- [Aluminum Boat Plans](#)
- [Marine Engineering](#)
- [Free Videos, E-Books & Reports](#)
- [Nautical Books](#)
- [Tips and Tricks Blog](#)

Home Boat Building - Lofting or Full Sized Plans?

I often get posed the question, "Do your plans come full sized or do you need to loft them?" I have to truthfully answer, neither, which often confuses people because they believe home built wooden boat plans are one or the other. In these days of computer aided design (CAD) drafting, no longer is it necessary to loft boat plans, because the computer actually does this for the boat designer. What this means to the boat builder, is that all they need to do is measure and mark, not loft.

Lofting is a process whereby the boat is actually drawn full sized to determine the exact sizes of the frames and components. Back when boat plans were drawn by hand in smaller scale, it was impossible to draw them accurately enough to ensure the finished boat had "fair" curves, that is to say, curves that smoothly transitioned from one part of the hull to the other. In the old days, a table of offsets or a half hull model was used to lay out several key points on the hull, and a long flexible piece of wood, called a spline was used to create a fair curve. From this full-sized drawing, measurements were taken to determine the exact sizes of the framing elements.

Enter the computer and Computer Aided Drafting (CAD.) This revolution in the boat designer's world allowed the naval architect to use an infinitely large piece of paper on which to design his boat full-size. Some of the early versions of AutoCAD came with a full scale drawing of the solar system with all of the planets and moons. It was 93 million miles from the earth to the sun! It gave you an appreciation of how huge the solar system actually is. On the other hand, I have offset lines one millionth of an inch and was able to zoom in until the two lines appeared on opposite sides of the computer screen.

So, no, you don't need to loft designs properly designed on CAD, and no there is no reason to need full sized plans or patterns, so long as you select modern plans by designers who use modern technology to design their boats. All plans from Spira International are computer generated for accuracy and pre-lofted in the computer.



Free Boatbuilding Information

I believe in offering you all the information you need to learn more about how to build a boat. That's why I offer these free videos, free ebooks, and free reports. You should know before you go, even if you decide on someone else's boat plans. I'm happy to help with whatever you need. Just email me or connect up using skype, social media, or any of the other free, modern methods if I can answer any questions you may have on boatbuilding.





•



•



•



•

Spira International, Inc., P.O Box 2155, Huntington Beach, CA 92647 - info@spirainternational.com - [Terms and Conditions](#)